

# *No Room For Them* - Composer's Notes

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This piece was composed in the autumn of 2025. It marks my first attempt at a symphonic arrangement, blending elements of classical music and New Age. The entire work is realized with a fully digital symphonic setup, requiring only three synthesizer keyboards and a small ensemble of four to five performers. By the pigeonhole principle, there will inevitably be at least one keyboard with more than two players standing in front of it.

## Fusion of Classical and New Age Styles

The advantage of adopting a classical foundation is that it allows for works with real depth and musicality — the creative ceiling is far higher than that of any popular music genre, a fact widely acknowledged in the industry. The New Age style, which emerged around 1980, brings two key benefits when fused into the mix:

1. **A richer sonic backdrop:** New Age makes heavy use of digital pads to paint the “background layer.” Not only does this blend seamlessly with traditional orchestral textures, but it also reduces the need for a 50-player string section.
2. **Structural flexibility:** Incorporating New Age elements does not compromise the ability to craft refined formal structures. A prime example is Yanni, whose signature work *Santorini* builds an entire 5-6 minute piece from a fixed motif based on a third interval — the same underlying logic Beethoven used in his *Hammerklavier Sonata* (Op. 106).

Although the style sits somewhere between classical and New Age, the instrumentation follows the “three-keyboard setup” that became popular after 2010 — a configuration originating in 21st-century pop music. The division of labor looks like this:

- **Primary melodic instruments:** handled by the first keyboard
- **Color instruments (2–3 types):** handled by the second keyboard
- **Textural layers (e.g., string pads):** handled by the third keyboard

Additionally, small percussion instruments should have a sense of spatial presence, which can be achieved by positioning players strategically on stage. Larger percussion can be performed live, triggered via synthesizer drum pads, or even pre-sampled and assigned to less frequently used keys — for instance, the highest or lowest octave of a keyboard.

# Instrumentation Setup

According to the commissioning party — Austin Chinese Church — two keyboards were already available:

- **Korg Kronos** (owned by the church)
- **Korg Kross** (personally owned by the band leader)

The church band also features excellent lead vocalists (both female and male), as well as a choir with many new members whose skills are still developing.

Among the two keyboards, the **Korg Kronos** stands out with what is arguably the best piano sound in the industry as of 2025 — so realistic that it can almost pass for a real concert grand. It’s the obvious choice for the **primary melodic instrument**. The **Korg Kross**, being an entry-level model, is better suited for **textural layers**, so the **string pad** part is assigned to it.

That left us missing one crucial piece: the **color instruments**. After consulting two composer friends and long-time collaborators — **Tao LIU** and **Yifan SHAN** — and reviewing professional evaluations, we all agreed that the **Yamaha MODX series** is the best choice. Its harp, clarinet, and trumpet patches are considered top of the industry (as of 2025), and these three are exactly what this piece requires as its primary color instruments. Conveniently, a used **Yamaha MODX8+** was available at the local Guitar Center here in Austin — so I quickly snapped it up.

With that, the essential “big three” are complete. The roles are assigned as follows:

| Instrument Category                                | Assigned To   | Reason                                                                                                      |
|----------------------------------------------------|---------------|-------------------------------------------------------------------------------------------------------------|
| Primary melodic instrument (piano) + Drum triggers | Korg Kronos   | Industry-leading piano sound. Touchscreen “drum pads” allow flexible percussion assignment.                 |
| Color instruments (clarinet + trumpet + harp)      | Yamaha MODX8+ | Industry-leading orchestral sounds. Three instruments fit perfectly, with two performers sharing the front. |
| Textural instruments (e.g., string pad)            | Korg Kross    | Doesn’t need to be perfect. Same brand as the main keyboard, ensuring seamless integration.                 |

# Orchestration

My overall approach is to follow, as closely as possible, the **fundamental logic of classical symphonic orchestration**:

- A well-balanced distribution of **high, mid, and low frequency ranges**
- A well-considered **left–right stereo placement**
- A carefully structured relationship between **point-like instruments** (e.g., harp) and **line-based instruments** (e.g., sustained strings)
- A clear and effective arrangement of **primary melody, secondary melody, texture, and harmonic foundation**

At the same time, certain **inconveniences inherent in classical orchestration** can now be addressed through **digital technology** — and this deserves special mention. Due to **physical acoustic limitations**, the **entire string section** (about **50–80 players**, including violins, violas, cellos, and double basses) produces roughly the **same volume as a single woodwind instrument** (such as a clarinet). Likewise, the **entire woodwind section** produces about the **same volume as a single brass instrument** (such as a trumpet).

This is extremely troublesome — I suspect **Beethoven, Brahms**, and others spent many sleepless nights grappling with it while composing their symphonies. Today, however, **digital technology allows us to boost the volume of individual instruments** freely and precisely.

Moreover, we can now **separate timbre from volume** — something that was impossible in the acoustic era. For instance, one of the most difficult tasks for singers is performing **high notes at a very soft dynamic (pp)** — something that is nearly impossible in traditional performance practice. But with digital tools, a singer can perform at a **slightly soft (mp) dynamic, step back from the microphone**, and have the **output volume reduced in post-processing** — and the result still sounds natural. For example, in this passage:

弱力度的高音，可离话筒远一些，以降低难度。  
May step back from the microphone to make weak high notes easier.

(Mezzo-)Soprano solo

57 *pp* 58 59 60 61 62

祈 愿 明 天 曙 光 升 起。\_\_\_\_\_

Therefore, my overall approach in this work is as follows:

1. The **woodwind section** is represented by the **clarinet**, chosen for its **exceptionally wide range**.

2. The **brass section** is represented by the **trumpet**, selected for its **superior projection and penetrating power**.
3. The **string section** is a bit more complex. **Sustained tones** are handled by a **string pad**, while **plucked (pizzicato) passages** are performed by the **piano** and **harp**. As for the **lyrical, singing-style string melodies**, current synthesizer keyboards (as of 2025) still struggle to reproduce them convincingly — they often sound rather **cheap and artificial**. For this reason, I decided to assign those lines to the **two vocal soloists** and the **clarinet** instead, thereby saving the manpower of **at least fifty string players**.
4. For the demo preview, I followed **Liu Tao's** suggestion and used **ACE Studio's AI vocalist**, which produced very impressive results.

### Technological Progress: Real-Time Preview

Traditional digital music workflows present a major inconvenience — especially for someone like me, who is accustomed to **writing directly in notation**. Once the score was written, I used to have to **export it, launch a separate program** (which could take half a minute to load), and go through a long, complicated **production process** before I could hear the rough result. This step was unavoidable: countless parameters had to be adjusted just to make the playback sound natural. It was tedious work — and, frankly, had **little relevance to the compositional phase itself**.

Even worse was the **recording process for vocalists**. If you're not a full-time professional composer, the time from **finding a singer** to **hearing the final result** can easily stretch to **one or two months** — by which time many original ideas are often **long forgotten**.

Fortunately, advances in technology have largely solved both problems. Since around **2020**, the notation software **Dorico** has made significant strides: as long as you **mark musical terms directly in the score** — for example, writing *pizz.* for pizzicato — Dorico will automatically **interpret and render the correct playing technique**. As a result, I no longer need to export the score, open another program, and wait endlessly — at least for the orchestral parts. I simply avoid **assigning lyrical string melodies** during composition, and the built-in playback is sufficient.

The same is true for the **“finding a vocalist”** process. Without a real vocal track, even an experienced composer can struggle to judge **how much instrumentation the accompaniment really needs**. A richly detailed vocal line allows the accompaniment to be much simpler — but this balance is notoriously hard to predict. Thanks to **AI vocal technology**, the entire process that once took **one or two months** can now be done in **five minutes** — a truly remarkable improvement.

In the original version, I orchestrated the **opening phrase of the soprano solo** with **piano, harp, and string pad chords**. But after switching to an **AI vocalist** (on the recommendation of **Tao LIU**), I realized that — based on the vocal detail — **piano ornamentation plus a single-note string pad** was more than sufficient:

S. solo. *mp* *p*  
走 过 山 岭 天 苍 茫, — 走

Pno. *p*

Hp *mp* *pp*

S.Pad. *pp* *p*

Mao Buyi's song *Sheng Xia (Midsummer)* follows a similar approach. However, before the advent of **AI vocalists**, I was never entirely sure whether this idea would truly work in practice — traditional **sample-library singers** were simply too poor in quality to serve as a reliable reference.

In addition, the **interaction between the piano and the harp** in this passage also works perfectly well:

Piano 70 71 72 73

70 71 72 73  
*p*

Harp 70 71 72 73  
*p*

The reason this works is that in the **digital music era**, we can **slightly reduce the distinct tonal character** of the piano and harp — for example, by **adjusting EQ** or **changing the instrument patches** — so that they blend more smoothly. We can even use **automation** to make these EQ settings **active only during these few measures**, with **fade-ins** and **fade-outs** applied to ensure a more seamless transition.

In the pre-digital era, if the piano and harp timbres simply didn't blend well, there was nothing one could do — you just had to **live with it**.

Additionally, in **measures 5–6**, there is a **color chord insertion** involving the **clarinet** together with the **piano** and **string pad** — a harmonic choice drawn from a **distant key relationship**.

The image shows a musical score excerpt for measures 4 through 7. The instruments are Clarinet (written in C) in Bb, Piano, Harp, and String Pad. The tempo is marked as 4/4 (q, 1+1+1+1) rit. The score includes various musical notations such as notes, rests, and dynamics (p, mp, p Red., mp Red.).

If the chord is played **entirely on the piano**, the effect is excellent. But when it's **distributed across multiple instruments** — especially when a **mark tree** (essentially a long row of wind chimes) is added — the result is noticeably less effective. The reason is that when a **single chord** is split among instruments with **significantly different timbres**, its **harmonic function and color** often become **diluted**.

On the piano, it typically takes about **0.3 seconds** for the listener to perceive a chord clearly. But when the chord is divided among timbrally distant instruments, it may take **0.6 seconds** or more. In measure **5**, the rapid series of **eighth notes** simply moves **too quickly** for the ear to register the harmony in time.

## Frequency Range and Beat Placement

For **pad-type instruments**, it's often effective to **enter on a weak beat** if possible. For example, in the score excerpt below, I've deliberately hidden the instrument names — but the **high sustained B** at the bottom is obviously a **string pad**, and its entrance on the **second beat** (a weak beat) works beautifully.

The **glockenspiel**, however, behaves differently. Its sound is **too prominent**, and if it enters on a weak beat **without a clear compositional purpose**, the result can feel **jarring** to the listener. In the example below, the glockenspiel **strikes on the second beat** starting in the

**following measure**, but it doesn't feel abrupt because the **string pad had already entered on the second beat** in the previous measure — preparing the listener **psychologically** for that rhythmic placement.

**B** Adagio cantabile ♩ = 68

The musical score is for a piece titled "Adagio cantabile" with a tempo of 68 beats per minute. It is written in 3/4 time. The score consists of six systems of staves. The first system shows a string pad entering on the second beat. The second system features a vocal line with lyrics "祈 愿 今 夜 草 中" and a string pad. The third system shows a vocal line with triplets and a string pad. The fourth system shows a vocal line with triplets and a string pad. The fifth system shows a vocal line with triplets and a string pad. The sixth system shows a vocal line with triplets and a string pad.

In addition, **string pads** are typically arranged in an **open voicing**, and their **highest note** is often placed **significantly above** the highest note of the **lead vocal or main instrument**, adding an extra layer of “**sheen**” or **brilliance** to the primary line.

## Composition

Both the first and second themes appear in two contrasting forms — a **minor-mode version** and a **major-mode version**.

### First Theme

The **minor version** of the first theme is as follows:



It first appears in **Section A** (the *semi-exposition*), where it is sung alternately by **two solo vocalists**. There is another instance of the **minor-mode version**, which we will discuss later.

The **major-mode version** of the first theme appears twice — both times in the **introduction**:



It is quite clear that the **first phrase** has a **4/4 feel**, while the **second phrase** (presented an octave lower) shifts into a **3/4 feel**. This subtly foreshadows the transition into **triple meter** in Section B and adds a touch of playfulness to the music. However, it also makes the clarinet “**response phrase**” that follows more challenging — the performer will likely need several attempts to confidently pinpoint the correct beat for the entrance.

In the semi-recapitulation (*Section C*):



It is quite clear that the note on “山” in measure **94** has been **stretched from its original one beat to three beats** — a significant contrast to how it was presented in the **semi-exposition (Section A)** in minor mode. To ensure the audience isn’t caught off guard by this change, I first **extended the note to two beats in Section B** as a kind of subtle “preview.”



To further reinforce the idea, the **female vocalist** repeats the phrase once more — this time in the **relative major key**.



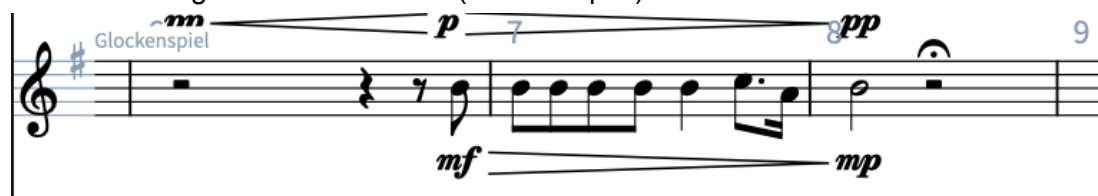


Since the **high note of the first theme** had already been **extended from one beat to two beats in Section B**, stretching it further to **three beats in Section C** no longer feels unexpected. As a result, musicians and singers with a good sense of musical structure — including several I’ve spoken with — can still immediately recognize it as the **first theme**, which is precisely what I intended.

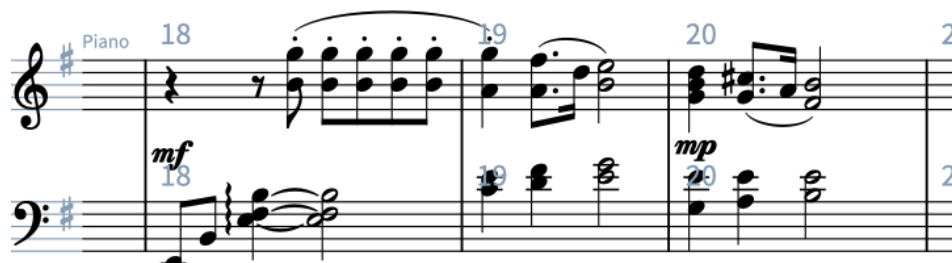
## Second Theme

The **minor-mode version of the second theme** is thoroughly **foreshadowed** both in the **introduction** and in **Section A (the semi-exposition)**.

Foreshadowing in the Introduction (Glockenspiel):



Foreshadowing in Section A (Piano):



Therefore, when the **second theme** reappears in **major mode** and is performed by the **choir in Section C (the semi-recapitulation)**, experienced listeners should feel a **joyful sense of both familiarity and freshness**:

Soprano 84 *pp* 85 86 87  
 马利亚约瑟 莫要迷茫, 待长夜散去 将有光。  
 Mezzo-soprano 84 85 86 87  
 马利亚约瑟 莫要迷茫, 待长夜散去 将有光。  
 Alto 84 *pp* 85 86 87  
 马利亚约瑟 莫要迷茫, 待长夜散去 将有光。  
 Tenor 84 *pp* 85 86 87  
 马利亚约瑟 莫要迷茫, 待长夜散去 将有光。  
 Bass 84 *pp* 85 86 87  
 马利亚约瑟 莫要迷茫, 待长夜散去 将有光。

## Combination of the First and Second Themes

In the **first half of Section C (the semi-recapitulation)**, the **second theme** is presented first — sung by the **choir** portraying *angels* — followed by the **first theme** reappearing in **major mode** performed by the **two lead vocalists**. As the section progresses, the two themes become increasingly intertwined and accelerated, eventually modulating from **B major** back to the **home key of E major**, pushing the music toward its climax.

As long as both themes are **fully established** and **prepared contrapuntally** in advance, combining them is not particularly difficult. However, attention must also be paid to the **lyrics**, which should evolve from “**independent statements**” to “**mutual responses**”, and finally to “**a unified message**.” Ideally, the audience should only hear **one line of text at a time**; occasional moments where “two groups sing different lines simultaneously” are acceptable, but this must not become the norm. After all, even three people speaking at once is confusing — let alone singing. Therefore, the real challenge lies in **careful structural planning**, ensuring the **music leaves sufficient space** for the lyrics to breathe.

Additionally, in an “**accelerando toward climax**” context, it becomes easy — and sometimes desirable — to write **non-square phrases** (i.e., phrases of 7 or 9 measures instead of the standard 8). A composer must be acutely aware of such **non-square structures**, understand their **purpose**, and **test their musical logic** repeatedly with *inner hearing*. Otherwise, the result will resemble the kind of *accidental 9-measure phrases* frequently seen in certain low-level worship music — where the composer had no clear intent, the performers keep missing entrances, and experienced listeners feel something is “off.” **Uncontrolled non-square structure** is a compositional mistake, often found among amateur writers.

My friend and fellow composer **Tao LIU** — whose credits include major films such as *Forever Young*, and *The Wandering Earth* (for which his original score — including its fundamental musical architecture and conceptual design — was appropriated and “rewritten” by another party after the producers maliciously withheld payment) — corrected me on this very issue **twice back in 2009**, after which I never made the mistake again.

## On Non-Square Structures

Let us take a closer look at **non-square phrase structures**. When **intentional and well-controlled**, they can make a musical sentence more **dynamic and expressive**. But when **unintentional and uncontrolled**, they should be **avoided at all costs**. A good composer must *always* **count the measures** to ensure they are not inadvertently writing a 7- or 9-measure phrase. Non-square structures must be **controlled**, not accidental.

Here are a few positive examples:



Take the **first theme** as an example. Its presentation in the **tenor solo** spans **nine measures**, from **measure 25 to 33**. The first two notes — on the word “*zou*” (走) — function as a **pickup** (*anacrusis*), and therefore are **not counted**. (In fact, I deliberately used a technique I call “**blurring the boundary between pickup and downbeat**” to *sneak in* an extra measure and create a **non-square structure** — for example, at the beginning of Section B.) In addition, the last syllable “*shang*” (上) in measure **34** overlaps with the beginning of the next section and is also **not included** in the count.

Thus, the **first theme** is a **non-square phrase**, but it doesn’t sound awkward. Instead, it conveys a sense of **relaxation** and **rhythmic elasticity**. This is because the irregularity stems from the word “*li*” (里) in measure **30** from the phrase “*returning home*” (*zhong gui gu li* / 终归故里). That single syllable is stretched by **two extra beats**. If you don’t believe me, try shortening “*li*” by two beats and shifting the following word “*ci chu*” (此处) forward by two beats — it will still sound completely natural.

There are **three reasons** why “*li*” can be stretched like this:

1. **Harmonic shift:** The harmony suddenly modulates to **A major (the tonic chord)**, and the listener needs a moment to *process* this surprise.
2. **Poetic phrasing:** In the lyrics, the rhyme scheme typically follows a “**1-2-4**” pattern (first, second, and fourth lines rhyme), so the **third line** — which contains “*li*” — naturally feels like a point for a brief breath.
3. **Melodic contour:** “*Li*” is a **third tone** and happens to be the **lowest point of the melodic contour** at that moment, so a slight pause feels musically appropriate.

This example illustrates a case where I **want the audience to notice the non-square structure** — to *feel* the sense of **stretch and release** it creates.

The next example is the complete opposite: here, I **don't want the audience to notice** the presence of a non-square structure:

The image shows a musical score for measures 48 to 57. The score is written for a vocal ensemble (Tenor solo, Soprano, Mezzo soprano, Alto, Tenor, Bass) and piano accompaniment. The key signature is one sharp (F#). The time signature changes from 4/4 to 3/4 at measure 48. The lyrics are in Chinese: 折 愿 今 夜 草 中 安 息. The piano part features a complex rhythmic pattern with triplets and eighth notes. The vocal parts are mostly rests, with some notes in measures 53-57.

Measures **48–52** form a short “**pre-introduction**” of **five measures** — clearly a **non-square structure**, much like the previous 9-measure example. But here, my **goal is the opposite**: I want the audience **not to notice** the irregularity. To achieve that, I used several techniques:

1. **Metric shift**: Starting at measure **48**, the meter changes from **4/4** to **3/4**. Strictly speaking, this makes measure 48 **ambiguous** — it can easily be perceived as belonging to the **previous phrase** (measure 47 and before).
  - If we group measure 48 with the previous phrase, that phrase itself becomes a **7-measure non-square structure**.
  - In other words, depending on how you divide it, the two phrases can be interpreted as either **6 + 5 measures** or **7 + 4 measures**.
2. **Beat reinterpretation**: The **high G** in measure **49**, if understood as the functional equivalent of the G in measure 48, should be interpreted as the **final beat of the previous bar**, not the **downbeat of the new one**. This deliberate **ambiguity “resets” the listener’s internal sense of meter**, making the irregularity far less noticeable.
3. **Total beat consistency**: In **4/4**, four measures add up to **16 beats**. After the switch to **3/4**, five measures total **15 beats** — nearly the same. This subtle consistency helps the passage **feel smooth and natural** despite the irregular phrase length.

4. The “**Rule of Three**”: Measures **49–52** follow the so-called “**rule of three**” — a structural principle not only found in **Propp’s *Morphology of the Folktale*** but also common across *temporal arts* like music. A short phrase is repeated **three times**: the first two are **unchanged**, while the third introduces a **noticeable variation**, leading into new material. In the score, **measures 50–52** feature a **glockenspiel strike on high B on the second beat**, subtly signaling that “*something is about to happen.*”
5. Precisely because I had already created a **five-measure non-square phrase** in measures **48–52**, introducing **another five-measure non-square phrase** immediately afterward in measures **53–57** no longer feels abrupt. On the contrary, since the audience can now perceive the irregularity, it produces a heightened sense of **stretch and release**.

## Modulation – Main Direction #1

The **first half of Section B** extends up to **measure 65**. At **measure 65**, there are **two possible overall modulation directions**. The **first direction** is the one adopted in the **current version**:

The musical score shows two vocal parts: Mezzo/Soprano (top) and Tenor (bottom). The key signature changes from B minor (one sharp) to C# minor (two sharps) at measure 63. The lyrics are: 我们彼此未曾离弃。 (We have never parted from each other.) and 同赞美：风 雨 彩 虹 是 神 奇 迹。 (Praising together: Wind, rain, rainbow, miracle of God.)

This version can be summarized as follows:

- At the **end of measure 61**, the music begins in **B minor**.
- By the **end of measure 63**, it **shifts up a whole tone** to **C# minor**. However, instead of continuing with strict sequential modulation, the harmony changes at “*qi*” (弃) in **measure 65**.
- As a result, by the **end of measure 65**, the harmony does **not** move up by another whole tone. Instead, it **jumps directly to G# minor**.

This creates the classic **III – IV – I** progression: **B – C# – G#**.

After briefly settling in **G# minor**, we again reach a fork in the road at **measure 69**. Here, too, there are **two main modulation directions**. The **first** is the one used in the **current version**, which modulates to the **relative major key — B major**:

67

Cl. in Bb

Glock.

S. solo.

T. solo.

Pno

Hp

风 雨 彩 虹 是 神 奇 迹。

风 雨 彩 虹 奇 迹。

*mf* *mp* *mp*

The **key point** here is **measure 69**. By taking advantage of the many **common chords** shared between **G# minor** and **B major** in the preceding measures, the music **directly** introduces the dominant chord of **B major** at this point.

However, it is also possible to **remain in G# minor** instead. To do this, you would simply **replace the chord** in **measure 69** with the dominant of **G# minor** and **slightly adjust the vocal lines** for the two soloists:

67 68 69 70 71

Cl. in Bb

Glock.

S. solo.

T. solo.

Pno

Hp

S.Pad.

风 雨 彩 虹 是 神 奇 迹。

风 雨 彩 虹 奇 迹。

*mf* *mp* *mp* *p*

A-Flat Major...

Adagietto cantabile ♩ = 80

Since **G# minor is enharmonically equivalent to A b minor**, and because the music later modulates to **A b major**, I chose to **notate from measure 69 onward in A b major** for the sake of readability.

This decision does **not diminish the artistic value** of the section in any way. I have also uploaded an MP3 version of this **alternative approach** to GitHub under the filename **Alternative1.mp3**, so you can listen and compare the difference for yourself.

## Modulation – Main Direction #2

As mentioned earlier, there are **two possible main modulation directions at measure 65**. The **first** is the one we have already discussed — the version used in the **current arrangement**. But there is also a **second possibility**.

In the first approach, the sequence is **B – C# – G#**, that is, a **III – IV – I** progression. This happens because I introduced a **chord change at the end of C#**, preventing the harmony from following the “expected” stepwise sequence **B – C# – D#**. Instead, it jumps directly to **G#**.

This naturally raises a question: *What if I don't make that chord change and simply let the music move “naturally” from C# to D#?*

The answer is: **of course I can**. And note that **D# is enharmonically equivalent to E b**. This leads us to the **second main direction**, which is structured as follows:

The image displays a musical score for measures 62 through 66. The score is written for four parts: S. solo (Soprano solo), T. solo (Tenor solo), Hp (Harp), and S. Pad. (String Pad). The key signature changes from one sharp (F#) to three flats (Bb, Eb, Ab) at measure 65, indicated by a red box labeled "E Flat Minor". The tempo is marked "Adagietto cantabile" with a quarter note equal to 74 (♩ = 74). The dynamics range from *pp* (pianissimo) to *mf* (mezzo-forte). The lyrics are in Chinese: "我们彼此未曾离弃。同赞美：苦难黑暗来袭，我们同赞：". The score shows a smooth modulation from C# minor to E-flat minor, with the harp and string pad providing a harmonic foundation.

67 68 69 70 71 72

Adagietto cantabile  $\text{♩} = 80$

Cl. in Bb

Glock.

S. solo.

T. solo.

Pno

Hp

S. Pad.

风 雨 彩 虹 是 神 奇 迹。

Similarly, I've uploaded an MP3 of this **alternative version** to GitHub under the filename **Alternative2.mp3**, so you can listen and compare.

Please note that **within the framework of “Main Direction #2”**, there are still **two possible sub-options** at measure 69: you can either **remain in E  $\flat$  minor** or **modulate to its relative major, G  $\flat$  major**.

### Modulation – Final Choice

By the end of **Section B**, combining the **two main modulation directions** discussed above yields a total of **four possible outcomes**. Starting from **E minor**, the four options are as follows:

1. **E minor** → **B minor** → (no modulation) → **B major**
2. **E minor** → **B minor** → (no modulation) → **A  $\flat$  major**
3. **E minor** → **B minor** → **E  $\flat$  minor** → **A  $\flat$  major**
4. **E minor** → **B minor** → **E  $\flat$  minor** → **G  $\flat$  major**

The advantage of “**opening two doors**” in Section B is precisely this: since **Section C** involves a significant **vocal workload** — and since vocal lines are subject to **range limitations** — not all keys are equally suitable. By having **four choices**, relatively evenly distributed across an **octave**, we gain tremendous **flexibility and room for adjustment** for the vocal parts.



Among these, the **first option** — the one used in the current version — is by far the most **logical and musically satisfying**. After entering **Section C**, the **B major** register falls squarely within the **comfortable and resonant range** for the human voice. In particular, the **soprano section of the choir** sings between **B4 and E5**, which is their **core tessitura** — a range that offers **excellent resonance, beautifully balanced tone quality**, and good **dynamic control** (especially important given that the choral dynamics in Section C build gradually from **pp** to **ff**). It is also worth noting that the **soprano soloist** eventually reaches a **high A (A5)** before modulating back to **E major**. On one hand, this **A5** serves as a **highlight pitch** for the soprano voice — placing it at the emotional and technical climax of the piece, something professional singers greatly appreciate. On the other hand, since the entire work is written in **E minor**, concluding in **E major** is musically natural and preserves **tonal unity** throughout the composition.

The **second option** is also worth considering — particularly if no soprano is available and the lead must be sung by a **mezzo-soprano** or **alto**. Because this option is a **minor third lower** than the first, the **highest note** drops from **A5 to F#5**, which is ideal for mezzos and still reachable for most altos.

The **third option** is more of a **bold, unconventional approach**. The reason is that if **Section C** begins in **E ♭ major**, that key lies exactly a **semitone away** from the overall tonal center of the piece (**E minor**). In such a case, it would feel more natural for **Section C** to modulate from **E ♭ major back to E major**, since this would effectively treat **E ♭** as a kind of “**prolonged leading tone**.” Structurally speaking, that reasoning is sound. However, a **semitone modulation** is inherently a **risky move** — much like using an “**unusual rhyme**” (*xian yun*, or “daring rhyme”) in poetry. If handled skillfully, the result can be extremely striking. My impression, though, is that this kind of semitone modulation doesn’t suit a “*hard pivot*” approach as often used in pop music. It works better if approached through **sequential motion** — but achieving such a sequence across a semitone gap is technically challenging. At the time, I didn’t have enough time to explore this path, so I decided not to pursue it. Perhaps I’ll revisit the idea in the future.

The **fourth option** is again a viable alternative, for similar reasons as the second. Because it is only **one tone lower** than the second, if the lead is sung by an **alto**, this approach works well — the highest note becomes **E5**, a **highlight pitch** for altos: rich, resonant, and warm, yet never harsh.

Roughly speaking, among the female vocalists I know, **Duan WANG** (soprano soloist of the original **Chongwenmen Church Women’s Choir**) is perfectly suited to the **first option**, while **Huimin LI** (alto soloist of the **China National Opera**) is an excellent fit for **options two and four**.

Since there is **one final modulation** to be made in Section C, the resulting key schemes become:

1. **E minor → B minor → (no modulation) → B major → E major** (*return to tonic – recommended*)
2. **E minor → B minor → (no modulation) → A ♭ major → D ♭ major**

3. **E minor** → **B minor** → **E $\flat$  minor** → **A $\flat$  major** → **A $\flat$  major**
4. **E minor** → **B minor** → **E $\flat$  minor** → **G $\flat$  major** → **B major**

Among these, **only the first option** naturally returns to **E major**. In options **2** and **4**, returning to **E major** would require **special handling in Section C**. Option **4** is particularly challenging (since **G $\flat$**  is a whole tone away from **E**), while option **2** is more manageable and potentially quite effective (because **A $\flat$**  and **E** are separated by a **major third**, making modulation easier and musically attractive). However, this approach sacrifices the “**long dominant pedal**” function that **B major** provides in option 1. Most listeners may not consciously notice the difference — but those with **strong musical training** (or **composer’s OCD tendencies**) will likely pick up on it and feel a subtle sense of regret.

## Formal Structure

This work adopts a “**semi-sonata form**” (*semi-sonata-allegro form*) in **E minor**, with the following sections:

- **Introduction (measures 1–8):**
  - Foreshadows the **first theme** (in **G major**)
  - Foreshadows the **second theme** (in **E minor**)
  - Introduces the overarching harmonic concept of **major–minor alternation** (see above)
  - Previews the local harmonic technique of **color-chord insertion** (measures 5–6)
  - Hints at the rhythmic interplay of **4/4 and 3/4 meters** (first 4 measures)
  - Introduces all major instruments, including the **trumpet**
- **Section A – Semi-Exposition:**
  - Presents the **first theme** (in **E minor**)
  - Provides a further preview of the **second theme** (in **E minor**)
  - Establishes a “**false dominant pedal**” (measures 42–46 on **G**) to create tension
- **Section B – Semi-Development:**

- Develops the **first theme** in various forms (including unconventional **3/4** treatment)
- Creates a **melancholic atmosphere** (drawing from **New Age** techniques)
- Concludes **in the dominant key (B major)**, foreshadowing a return to the tonic (E minor / E major)
- **Section C – Semi-Recapitulation:**
  - Presents the **second theme** in **major**, forming a stark contrast with its earlier minor appearance
  - Presents the **first theme** in **major**, again contrasting its earlier minor version
  - Together, these serve as an **extended dominant pedal** (measures 84–108), building **potential energy** for the final return to the tonic (E)
  - Concludes in **E major**

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The term “**semi-sonata form**” is used because each section fulfills **half of the traditional sonata functions**, albeit through alternative means.

For instance, **Section A (semi-exposition)** presents the **first theme** and creates tension through a **dominant pedal** (measures 42–46). However, the **second theme** is only **previewed**, not fully presented, and the dominant pedal does not function as a bridge leading to it (since the second theme itself never arrives here). Instead, it serves as a bridge toward the **semi-development** section.

**Section B (semi-development)** follows the same principle. It remains in **E minor** rather than modulating to a new key, because the overall length of the work is not sufficient to support frequent **tonal center shifts**. Furthermore, the development section does not end with a dominant pedal (i.e., it is not “open”) but rather **resolves into the dominant key (B major)**. This is because the full presentation of the **second theme** (as opposed to its earlier previews) is performed by the **choir**, portraying the **voices of angels** — a moment that requires silence and attentiveness from the audience. In such a context, a “**closure–restart**” structure is far more appropriate than the traditional “**suspense–resolution**” trajectory of sonata form.

**Section C (semi-recapitulation)** follows a similar logic. In classical sonata form, the development is usually followed by a **large dominant pedal** to build tension before the recapitulation. Here, however, the **first half of Section C** is presented in the **dominant key (B major)**, which itself serves as an **extended dominant pedal**, accumulating enough **energetic momentum** for the final **return to E major** at the conclusion.

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